SHRI RAMSWAROOP MEMORIAL UNIVERSITY

End Semester Examination (2024-25)–Odd Semester

B. Tech (CSE) (DS & AI)/ B. Tech (CSE) (CC & AI)/B. Tech (CS) (Artifical Intelligence and Machine learning Engineering) - I year (I Sem)	
Course Name: Chemistry	Code: UCY1005
Time: 03 Hours	Max Marks: 60

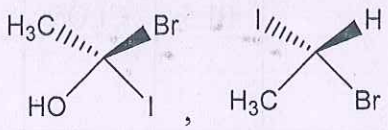
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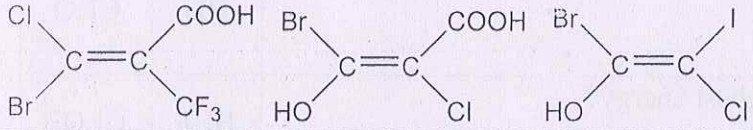
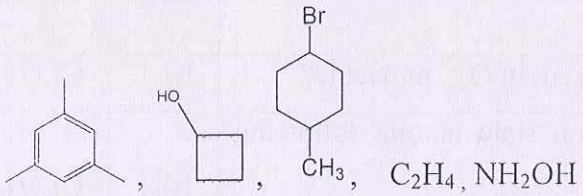
(To be filled by the Student)

Note: Please read instructions carefully:

a) *The question paper has 03 sections and it is compulsory to attempt all sections.*

b) *All questions of Section A are compulsory; questions in Section B and C contain choice.*

Section A: Very Short Answer type Questions **Attempt all the questions.**		**BL**	**CLO**	**Marks (10)**
1.	Which of the following radiation has highest energy? Radio wave, Microwave, Infrared radiation.	BL1	CLO2	01
2.	List the criteria for a molecule to be aromatic.	BL1	CLO1	01
3.	Explain intra- molecular hydrogen bonding.	BL2	CLO3	01
4.	Discuss equivalent and non- equivalent proton with suitable examples.	BL2	CLO2	01
5.	Explain the term entropy. How entropy relate with enthalpy and Gibb's energy?	BL2	CLO3	01
6.	How many number of unpaired electron is/are present in O_2^{2-} molecule?	BL1	CLO1	01
7.	Identify the element which has highest oxidation state among following molecule? H_2SO_4, $HClO_4$, HNO_3	BL1	CLO4	01
8.	Discuss the term electrophile with suitable examples.	BL2	CLO5	01
9.	Which type of salt is/are responsible for permanent hardness in water?	BL1	CLO4	01
10.	Find R/S configuration of given compounds;  	BL1	CLO5	01
Section B: Short Answer Type Questions **Attempt any 08 out of 09 questions.**		**BL**	**CLO**	**Marks (20)**
1.	Classify the acids and bases in terms of Bronsted – Lowry concept? Give suitable example.	BL2	CLO4	2.5
2.	Illustrate the degree of freedom. Calculate the number of vibration mode for NH_3 and C_2H_2 molecule.	BL3	CLO2	2.5

3.	Distinguish between ionic and covalent radii. Arrange the following in increasing order of their atomic size; Mg^{2+} , Na^{+}, F^{-}, Ne.	BL4	CLO4	2.5
4.	Show the molecular orbital diagram of 1.3-butadiene and identify HOMO and LUMO orbital.	BL3	CLO1	2.5
5.	Explain Nernst equation for electromotive force (EMF).	BL2	CLO3	2.5
6.	Calculate the energy difference between n=3 to n=4 for a particle in 1-D box.	BL3	CLO1	2.5
7.	Outline the structure of paracetamol and aspirin. Define medicinal use of these two.	BL4	CLO5	2.5
8.	Illustrate the selection rule for a molecule to be microwave active. Out of CO_2, NO, H_2O, H_2 and N_2,which will exhibit pure rotational spectra.	BL4	CLO2	2.5
9.	Illustrate the term stereoisomer and predict the E/Z configuration of following compounds and define the term isomer. 	BL3	CLO5	2.5
Section C: Long Answer Type Questions **Attempt any 05 out of 06 questions.**		**BL**	**CLO**	**Marks (30)**
1.	Discuss the band theory and effect of temperature on conductivity of conductor, semi-conductor and insulator.	BL5	CLO1	06
2.	Discuss the term shielding and de-shielding and predict the number of H-NMR signal for following compounds;  , , , C_2H_4 , NH_2OH	BL6	CLO2	06
3.	Explain the following terms with suitable example; (a.) Oxidation and reduction process (b.) Dipole-dipole interaction (c.) Corrosion	BL5	CLO3	06
4.	Summarize the following terms with suitable examples: (a.) Enantiomer (b.) Diastereomer (c.) Chiral centre (d.) E_2-elimination reaction	BL5	CLO5	06
5.	Discuss the crystal field theory in detail and determine the CFSE value and magnetic moment of following complexes; (i) d^3 (ii) d^6 (low spin) (iii) d^4(high spin) (iv) d^9 (v) d^{10}	BL6	CLO1	06
6.	Discuss the Fazan's rule and identify the following species as hard acids and soft acids; H^+, Na^+, Cu^+, I^+,Ca^{2+},Ag^+	BL6	CLO4	06
